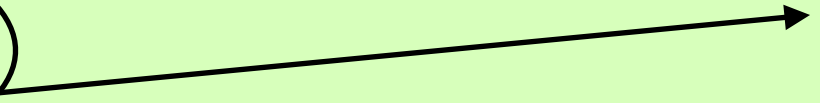




Welcome! 😊

EECS 376, Sec. 18
D4: More DP+Greedy

Sam Hiken



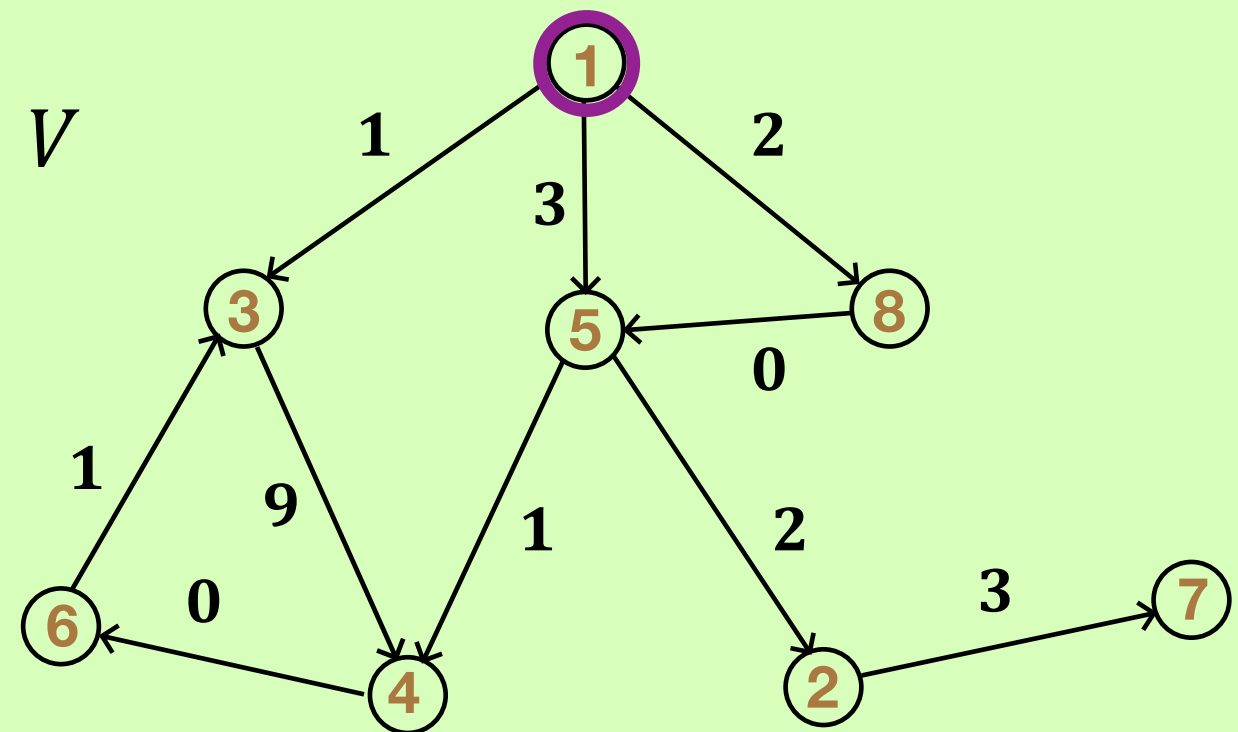
Dynamic programming on graphs

Single-source shortest paths:

- Input: $G = (V, E)$, source node $s \in V$
- Output: for every $t \in V$: $d(s, t)$

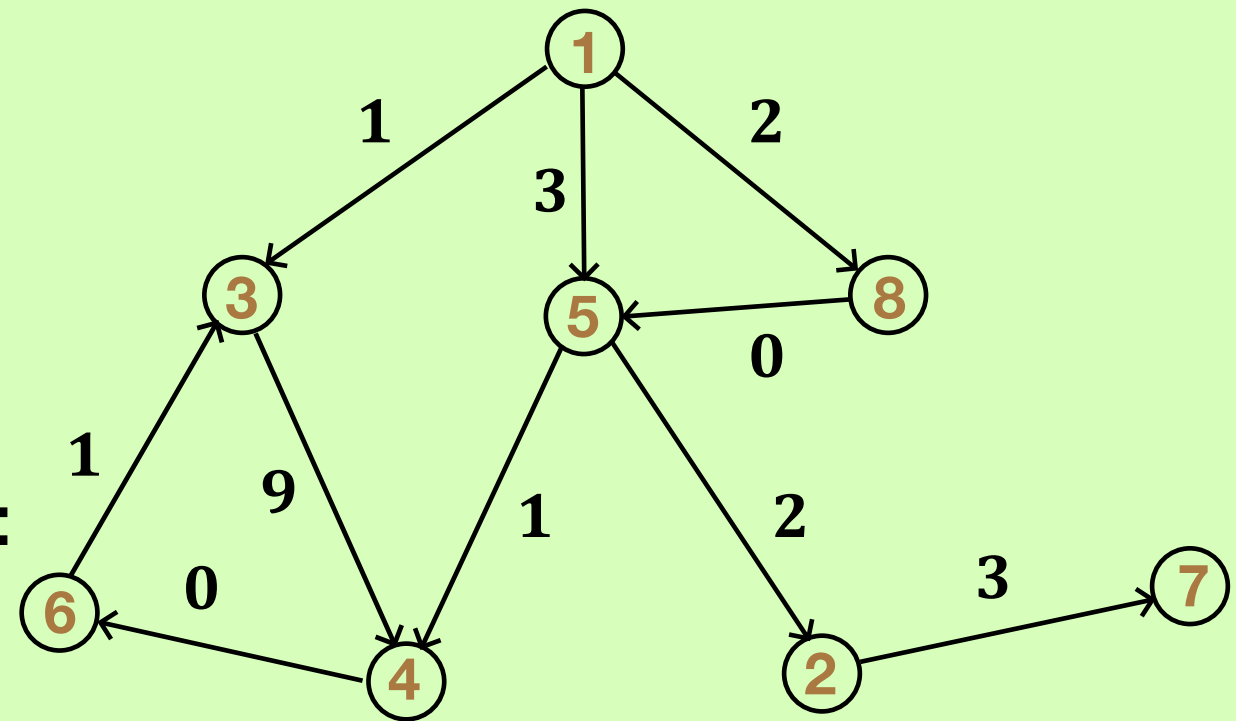
All-pairs shortest paths:

- Input: $G = (V, E)$
- Output: for every $s, t \in V$: $d(s, t)$



Bellman-Ford

For single-source shortest paths:



1	2	3	4	5	6	7	8
0	∞	∞	∞	∞	∞	∞	∞

0	∞	1	∞	3	∞	∞	2
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0	5	1	4	2	∞	∞	2
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0	?	1	3	2	4	8	2
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Quiz: what fills in the ?:

A) 5

B) 2

C) 4

Floyd-Warshall

All-pairs shortest paths

Runs in time $O(|V|^3)$.

Fills out a $|V| \times |V| \times |V|$ DP table

$T[k, i, j]$ = weight of the shortest $i j$ path using only vertices from $\{1, \dots, k\}$

Greedy algorithms

Makes choices that are **locally optimal**

Example: Kruskal's algorithm

Exchange argument: used to prove a greedy algo's output is **globally optimal**

Exchange argument

Used to prove that a greedy algorithm returns a **globally optimal** solution.

Premise:

Let OPT be globally optimal solution $\{s_1, \dots, s_m\}$

Let ALG be the output of a greedy algorithm

Repeatedly replace elements of OPT with elements from ALG

Show that each time, the cost of OPT does not increase.

Gather into groups of 2-3, work on the worksheet

In the last ~10 minutes, we will discuss the answers to **problems 2 and 4**

Link to the worksheet: <https://tinyurl.com/yyaeudas>
[this is in the class Google Drive]

Attendance link: [given in lecture hall]